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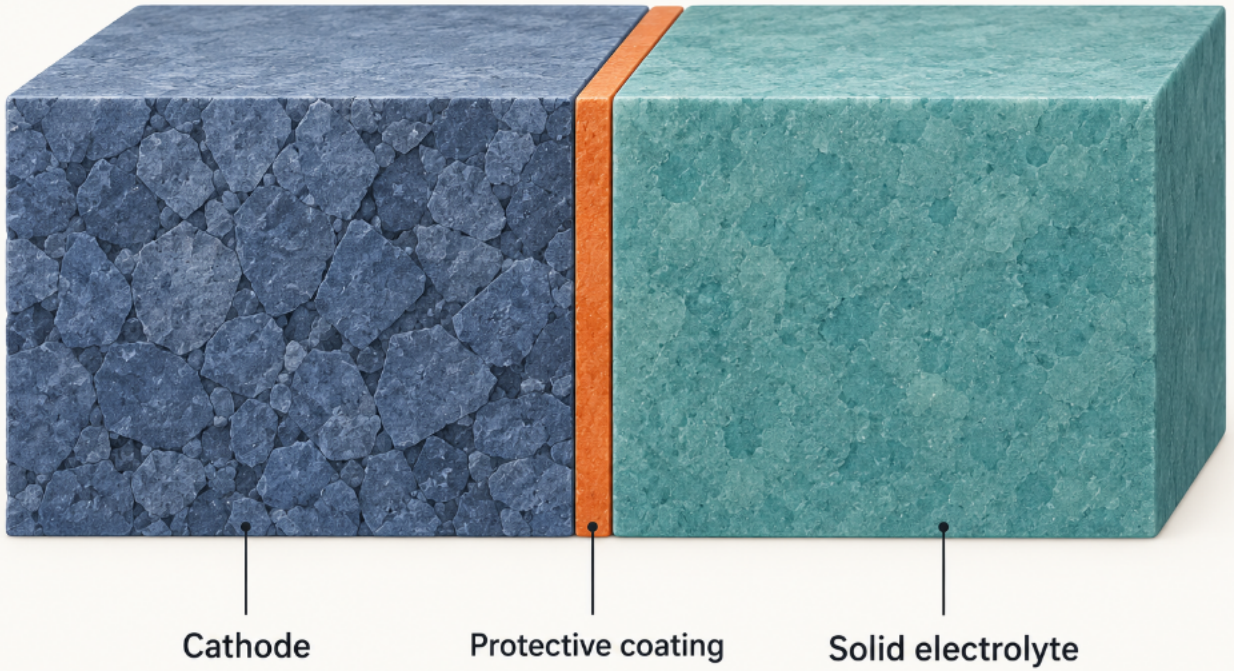
Can an LLM choose the next battery coating to test?

Why the LLM matters
What it achieved, step by step

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The problem: a protective layer inside a battery

The cathode stores lithium. The electrolyte lets lithium ions move.



Conceptual illustration, not to scale

A useful coating must

Limit unwanted reactions at both contacts.

Stay attached to the neighboring materials.

Lithium ions must also pass through. That was not tested.

Why the LLM is central

LLM = large language model. Here, it chooses what to calculate next.

01 **Connect chemistry to a candidate**

Choose a known material and explain why it might work.

02 **Use the last result**

Read what failed or nearly passed, then choose the next test.

03 **Make a prediction that can fail**

State an expected result before the physical calculation.

Researchers define the task. The LLM makes choices. The physics tools calculate the results.

One decision loop, repeated after each result

1 **LLM: choose a material**

Record the proposed coating and its chemical rationale.

2 **Tool: test the material**

Return a numerical result from the defined calculation.

3 **LLM: interpret the feedback**

Compare the result with the earlier expectation.

4 **LLM: choose the next test**

Continue until the evaluation budget is used.

First test: can it find a coating that passes?

Choose from a fixed menu of known materials.

A calculation scores unwanted chemical reactions.
A material passes only if it meets the preset limits.

10 tries

per search

30 searches

per strategy, with varied candidate order

“Success” means finding at least one passing material. It does not mean a working battery.

Five strategies make the comparison fair

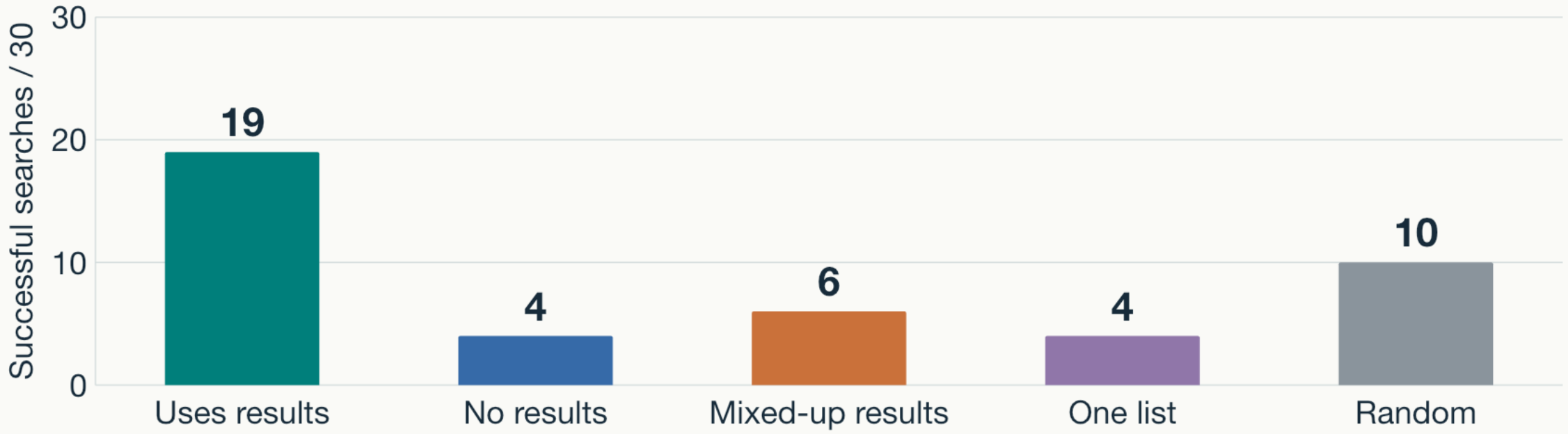
The same menu, the same ten tries, and the same definition of success.

Strategy used in the charts	How the next choice is made
Uses results	LLM sees the outcomes of its earlier choices
No results	LLM sees its choices, but not their outcomes
Mixed-up results	LLM sees outcomes reassigned between materials
One list	LLM chooses all ten materials before any result
Random	Software picks materials at random, without repeats

The first four use the same LLM. Random search provides a simple comparison.

Using results improved the first search test

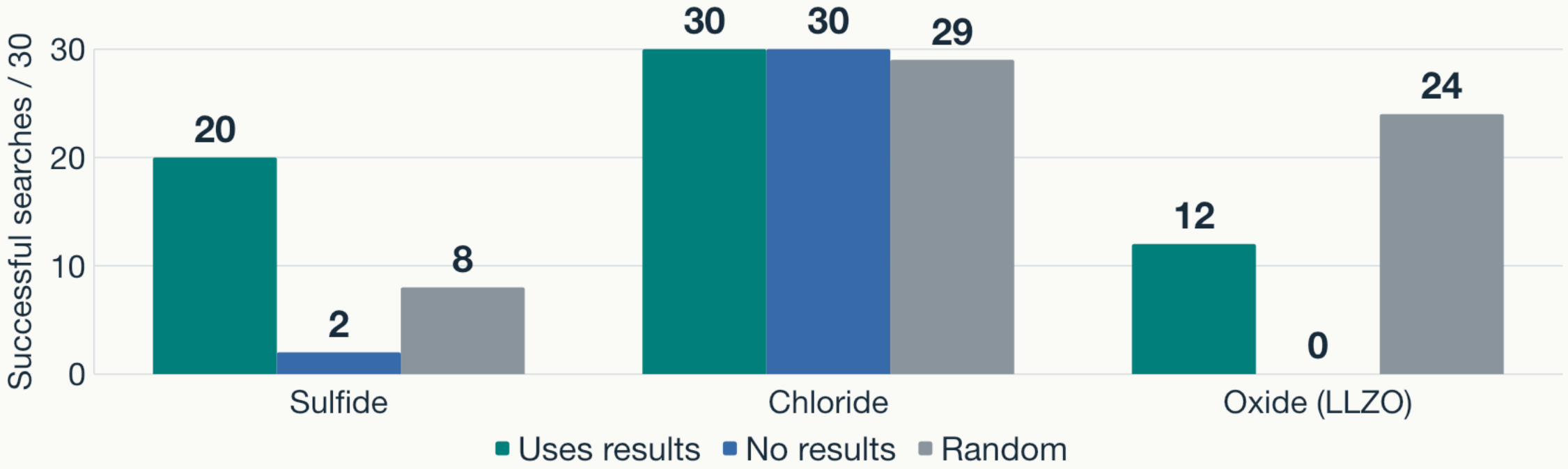
Successful searches out of 30, using the original 50-structure menu.



Correct feedback helped: 19 successes, versus 4 without results and 6 with mixed-up results.

The benefit changed with the battery materials

141 structures, three electrolytes, 450 searches across all five strategies.



Feedback helped

Almost all succeeded

Random did better

A real example: refining a near miss

Three consecutive choices in one sulfide search. Lower reaction score is better.

Try	Material chosen	Reaction score	What the test said
6	Li_3AlF_6	0.160	Does not pass
7	$LiMgAlF_6$	0.057	Close, but still above the limit
8	$LiCaAlF_6$	0.048	Passes the 0.050 limit

The LLM kept testing related chemistry after a promising near miss.

One illustrative search. Reaction scores are in eV per non-lithium atom.

On the oxide, the LLM tried fewer chemical options

LLZO is the oxide electrolyte in the previous chart.

33

different chemical formulas
tried by the LLM using results

95

different chemical formulas
tried by random search

10 of its 12 successful searches first passed on tries 8–10.

Across 300 choices per strategy. Broad exploration was more effective in this test.

Next test: predict how strongly the coating attaches

One known reference

The LLM sees calculated results for $LiNbF_6$, then chooses two more coatings from a small menu.

A prediction before the test

For each choice, it predicts adhesion at both contacts before the new results exist.

Adhesion = energy needed to separate a contact in this model.
Higher values mean stronger attachment under the tested conditions.

PET-MAD simulates the atoms. These results have not been measured in a real battery.

STEP 3 / PREDICTION CHECKED

First choice: the forecast worked for $LiTaF_6$

Its idea: replace niobium (Nb) with chemically related tantalum (Ta).

Contact with the coating	LLM forecast	Calculated adhesion
Cathode side	3.2–3.9	3.339 (within range)
Electrolyte side	1.4–1.9	1.577 (within range)

Both forecasts held. It also predicted the weaker contact correctly.

Units: J/m². Forecasts were recorded before the new calculations.

Second choice: the forecast failed for $LiCaAlF_6$

The LLM expected stronger attachment than the calculations found.

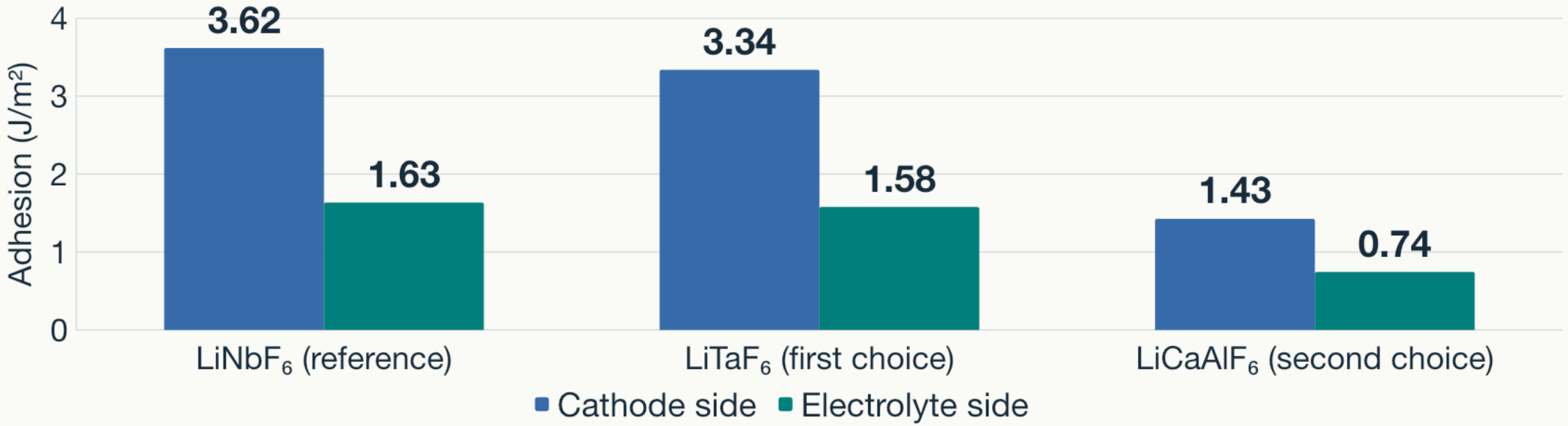
Contact with the coating	LLM forecast	Calculated adhesion
Cathode side	3.0–3.8	1.427 (below range)
Electrolyte side	1.4–1.9	0.744 (below range)

Both forecasts missed. The LLM acknowledged the failure afterward.

Units: J/m². It still identified the electrolyte side as the weaker contact.

What the physical calculations found

Adhesion in the tested structures. Higher values mean stronger attachment.



Ca–Al attached less strongly. The small Nb–Ta gap does not establish a reliable winner.

Numerical sensitivity and charged model surfaces still limit the physical interpretation.

What the LLM achieved

01 **Useful feedback improved some choices**

Strong gains in the sulfide tests. The oxide result showed a limit.

02 **It made chemical hypotheses testable**

Two adhesion forecasts held and two failed, all recorded in advance.

03 **It assessed results, but still needed review**

Its order stayed the same. A human corrected one interpretation.

We tested a scientific decision process. A better battery coating has not yet been validated.

What comes next

1 **Improve how the LLM explores**

Test ways to avoid repeating a narrow set of chemical ideas.

2 **Test choices over more physical calculations**

Compare LLM-guided selection with simple rules and random choices.

3 **Check the missing battery requirements**

Test ion transport, practical preparation and experimental performance.

The contribution: a measured test of AI-guided scientific choices, including where they fail.

All five strategies and the planned statistical tests

Electrolyte	Uses results	No results	Mixed up	One list	Random
Sulfide	20	2	6	0	8
Chloride	30	30	30	30	29
Oxide (LLZO)	12	0	4	1	24

Successful searches out of 30, after ten tries. Holm-adjusted paired tests:

Sulfide: versus no results p = 0.0000458, versus mixed-up p = 0.00244.
LLZO: versus no results p = 0.00244, versus mixed-up p = 0.172.
Chloride: p = 1 for both comparisons. Six planned tests in total.

Acceptability, optimum-finding and practicality differ

Question	Measure	Observed limitation
Can the policy find a passing candidate?	Acceptance at $P \leq 0.050$	This is the primary reported result
Does it find the pool's optimum?	Best-bin discovery / regret	Sulfide: iterative 6, random 5 LLZO: all LLMs 0, random 2 (of 30)
Is the actual coating useful?	Transport, processing, electrochemistry	Not established by this evaluator
Is the search saving compute?	Cost of avoided expensive evaluations	Cheap tables alone cannot show this

Acceptance cuts through a ranking bin. Near-threshold conclusions depend on the frozen definition.

The LLZO audit qualifies the criterion, not the score

5 of 17 acceptable structures were flagged for practical review: $LiIO_3$, $LiClO_4$ and Li_5IO_6.

Random's 24 successful runs	Count
Also found an unflagged acceptable candidate	18
Succeeded only through a flagged candidate	6

This is a descriptive partition, not a rerun with a new endpoint. Primary result: 24 versus 12.

The model is not proven to be making a safety judgment. Finite-temperature and processing limits remain.

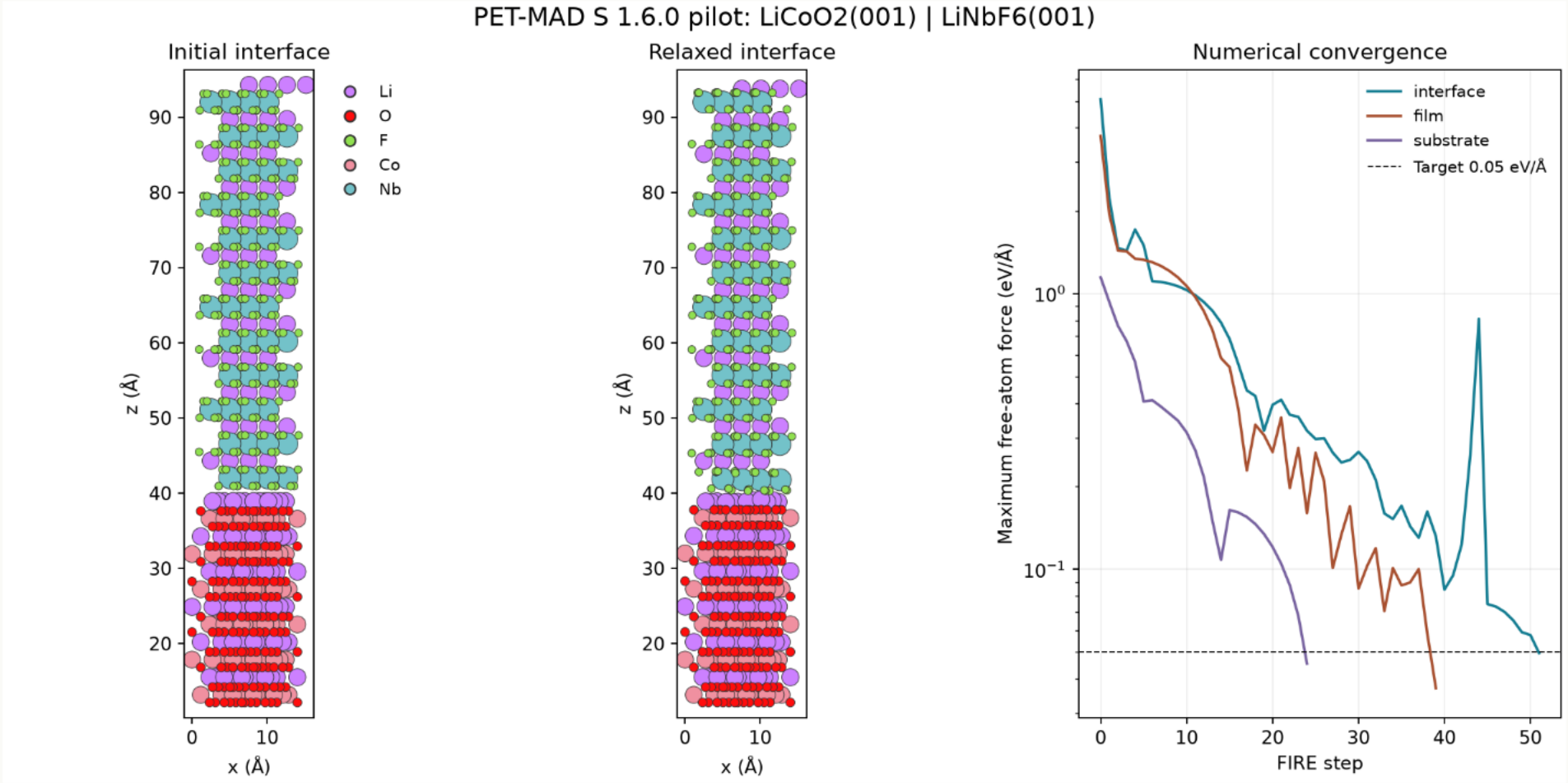
Relaxed adhesion and rigid separation are different

Coating	Cathode Wadh / Wsep	Electrolyte Wadh / Wsep	Atoms C / E
$LiNbF_6$	3.617 / 5.168	1.633 / 2.967	696 / 880
$LiTaF_6$	3.339 / 4.728	1.577 / 2.911	696 / 880
$LiCaAlF_6$	1.427 / 4.322	0.744 / 1.490	720 / 1212

Wadh: separated slabs are allowed to relax.
Wsep: fragments retain their joined-interface geometry.

J/m². Primary snapshots unchanged. Later robustness checks are a separate record.

The structures and convergence evidence are archived



Archived evidence and completed integration

Evidence / cost	Verified record
Environment study	450 runs, 2,800 API calls, about $3.63 at recorded list rates
Physical pilot LLM	5 / 10 HTTP attempts, about $0.006 at recorded list rates
New PET-MAD jobs	26 / 36, all converged, 6,632 runner calls + 52 single points
Compute time	25.4 min runner time; 47.5 min summed Slurm elapsed Reused work, bulk preparation and queue time are separate
Integrated delivery	Reader fix, code integration and structure viewer completed on the separate final-integration branch

Frozen decisions, later robustness, source hashes and limitations remain independently inspectable.